

## 1. What is Manual Testing?

**Manual Testing** is the process of testing software **manually by a human**, without using automation tools.

Tester checks whether the application works correctly by:

- clicking buttons
- entering data
- checking outputs
- finding bugs/issues

### Example:

Login page එකක් තියෙනවා.

ඔයා username/password type කරලා **Login** click කරනවා.

System එක expected විදිහට work කරනවද බලනවා.

That is **manual testing**.

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## 2. Why Manual Testing?

Manual testing helps to:

- Find bugs
  - Check if features work properly
  - Ensure good user experience
  - Verify requirements
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## 3. SDLC vs STLC

**SDLC = Software Development Life Cycle**

Software build කරන steps:

1. Requirement gathering
2. Design
3. Development

4. Testing
  5. Deployment
  6. Maintenance
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## **STLC = Software Testing Life Cycle**

Testing steps:

1. Requirement Analysis
  2. Test Planning
  3. Test Case Design
  4. Test Environment Setup
  5. Test Execution
  6. Test Closure
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## **4. Types of Manual Testing**

### **Functional Testing**

Check whether features work correctly.

Example:

- Login works?
  - Add to cart works?
  - Search works?
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### **UI Testing**

Check design and interface.

Example:

- Button visible?
- Text aligned?

- Colors correct?
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### **Usability Testing**

Check if system is easy to use.

Example:

- Navigation easy?
  - User understands buttons?
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### **Regression Testing**

Check old features after new changes.

Example:

Developer fixes login bug → tester checks login + other related pages.

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### **Smoke Testing**

Basic quick testing.

Example:

Application opens?

Login works?

Dashboard loads?

If smoke fails → no need further testing.

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### **Sanity Testing**

Quick check after small bug fix.

Example:

Password reset bug fixed → test only password reset.

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### **Exploratory Testing**

No predefined test cases.  
Tester explores app and finds bugs.

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## **Ad-hoc Testing**

Random testing without planning.

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## **5. Test Case**

A **Test Case** is a document describing what to test.

Example:

**Test Case ID**    **TC\_001**

Test Scenario    Login

Test Steps        Enter valid username and password

Expected Result   User logs in successfully

Actual Result      Passed

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## **6. Test Scenario**

High-level thing to test.

Example:

**Verify login functionality**

One scenario can have many test cases.

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## **7. Bug / Defect**

When actual result  $\neq$  expected result.

Example:

User enters correct password but cannot login.

That is a **bug**.

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## 8. Bug Life Cycle

Bug stages:

1. New
2. Assigned
3. Open
4. Fixed
5. Retest
6. Closed

Possible other states:

- Reopened
- Rejected
- Deferred

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## 9. Severity vs Priority

### Severity

How serious the bug is.

Example:

App crashes = High severity

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### Priority

How quickly it should be fixed.

Example:

Logo spelling mistake = Low severity but maybe high priority

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## 10. Test Plan

A document explaining:

- what to test
  - how to test
  - who tests
  - schedule
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## **11. Test Data**

Input values used for testing.

Examples:

- valid username
  - invalid password
  - empty fields
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## **12. Positive and Negative Testing**

### **Positive Testing**

Valid inputs.

Example:

Correct username/password.

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### **Negative Testing**

Invalid inputs.

Example:

Wrong password.

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## **13. Retesting**

Testing fixed bug again.

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## **14. Regression Testing**

Check whether fix affected other features.

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## **15. Real Example**

Testing login page:

Test cases:

1. Login with valid credentials
  2. Login with invalid password
  3. Login with empty username
  4. Login with empty password
  5. Password hidden?
  6. Forgot password works?
  7. Error message shown?
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## **Interview Questions**

### **What is manual testing?**

Manual testing is software testing performed manually without automation tools to identify defects.

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### **Difference between test case and test scenario?**

Test scenario = high-level idea

Test case = detailed steps

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### **Difference between severity and priority?**

Severity = impact of bug

Priority = urgency to fix

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### **What is regression testing?**

Testing existing features after changes to ensure nothing broke.

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### **What is smoke testing?**

Basic testing to check whether build is stable.